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Appendix B — What should just work, and how to verify it

This is the at-a-glance reference for a finished build: every URL, what it's for, and the one-line check that proves it's healthy. Use it as a smoke test after a reboot, after an update, or any time something feels off — work top to bottom and the first failing row tells you which layer broke.

Throughout, substitute your own values where this guide uses placeholders: the Pi's Tailscale IP for `100.x.y.z` (run `tailscale ip -4` on the Pi) and your tailnet suffix for `your-tailnet.ts.net`.

The one-time cloud setting everything depends on

Almost everything below is configured *on the Pi* and works the moment the containers are up. The **single exception** is a setting in the Tailscale admin console that can't be done from the Pi — and until it's on, the `.home` names and tailnet-wide ad-blocking won't work on your *other* devices:

! Make Pi-hole your tailnet's DNS (admin console, one time)

In the Tailscale admin console (<https://login.tailscale.com>) → **DNS** page:

1. **Add nameserver** → **Custom**, enter the Pi's **Tailscale** IP (`100.x.y.z`), leave *Restrict to domain* **off**, and *Use with exit node* **off**. Save.
2. Turn on **“Override local DNS.”**

This is the [Part 4 Step 5](#) step. Verify it took with `tailscale dns status` on any device — under *Resolvers* you should see your Pi's `100.x.y.z`, not “no resolvers configured.” If it still says no resolvers, the admin-console setting hasn't applied yet.

The service map

Service	URL (everywhere on your tailnet)	What it does
Dashboard (Homepage)	<code>https://home.home</code> (or <code>https://homelab</code>)	Landing page; live tiles + links
Audiobookshelf	<code>https://abs.home</code>	Stream your audiobooks / Assimil

Service	URL (everywhere on your tailnet)	What it does
Pi-hole	<code>https://pihole.home</code>	DNS ad-blocking admin
Portainer	<code>https://homelab:9443</code>	Docker management GUI

All four work from any device signed into your tailnet — home Wi-Fi or cellular — with no IPs and no port numbers (except Portainer, which keeps its own :9443).

Verify it, layer by layer

Run these in order. Each isolates one layer, so the first failure pinpoints the problem.

1. The containers are all up (on the Pi):

```
docker ps --format "table {{.Names}}\t{{.Status}}"
```

You should see five **Up** containers: `caddy`, `homepage`, `pihole`, `portainer`, `audiobookshelf`.

2. They share the proxy network (on the Pi):

```
docker network inspect homelab --format "{{range .Containers}}{{.Name}}\n{{end}}"
```

All five names should appear. If `pihole` or `audiobookshelf` is missing, the reverse proxy can't reach it — re-read the network callout in [Part 4](#).

3. DNS resolves the pretty names (on the Pi):

```
dig +short pihole.home @127.0.0.1      # expect your Pi's Tailscale IP (100.x.y.z)
dig +short doubleclick.net @127.0.0.1 # expect 0.0.0.0 (ad-blocking works)
```

4. The proxy routes each name to the right service (on the Pi):

```
for name in pihole.home abs.home home.home; do
  printf "%-12s -> " "$name"
  curl -s -o /dev/null -w "HTTP %{http_code}\n" \
    --resolve "$name:80:${tailscale ip -4 | head -1}" "http://$name/"
done
```

Expect `pihole.home` -> HTTP 302 (it redirects to `/admin`) and `abs.home` / `home.home` -> HTTP 200. This tests the full chain — DNS name → Caddy on the Tailscale IP → the correct backend container — exactly as a remote device sees it.

5. The names work from another device (laptop/phone on the tailnet):

```
ping home.home          # answers from the Pi's 100.x.y.z
```

Then open `https://home.home`, `https://abs.home`, and `https://pihole.home` in a browser. If `ping` fails here but step 4 passed on the Pi, the missing piece is the **Tailscale DNS override** (the cloud setting at the top of this appendix).

6. The dashboard widgets have live data:

Open `https://home.home`. The Pi-hole tile should show today’s blocked-query count and the Audiobookshelf tile your library — proof the widgets authenticated through the **homelab** network using the secrets in `~/dashboard/.env`. If a tile says “API error,” see the widget troubleshooting in [Part 5](#) (for Pi-hole, the usual culprit is a missing `version: 6`).

What works away from home (and what deliberately doesn’t)

Once the cloud setting above is on, here’s the honest picture the moment you leave the house — the full reasoning is in [Part 7](#):

Mode (away)	Homelab URLs?	Pi-hole ad-block?	Hides your IP?
Tailscale on, no exit node (everyday default)			
Tailscale + your Pi as exit node			(home IP)
Tailscale + Mullvad exit node		(Mullvad DNS)	
Aura (or any standalone VPN) on			

The one rule behind the table: a phone runs **one** VPN at a time, and whatever VPN is active owns DNS. So homelab access, Pi-hole filtering, and a privacy VPN can’t all be on at once — you switch between coherent modes. The everyday default (top row) is the one you live in.

Reboot behavior

Nothing extra is required after a power cycle. Every container uses a `restart: policy` (`unless-stopped` or `always`), so Docker brings them all back on boot; the **homelab** network is persistent; and the `systemd-resolved` stub stays disabled (the drop-in from [Part 3](#) survives reboots), so Pi-hole reclaims port 53 cleanly. Reboot the Pi and re-run the verification steps above — every row should pass without you touching anything.